

Lynbrook Math Summer Camp 2026

A4 Sequences and Series

Lynbrook Math Club

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A4.1 Lecture

Definition A4.1.1. A **sequence** is an ordered list of terms.

Definition A4.1.2. A **series** is the sum of the terms in a sequence.

Arithmetic and Geometric Sequences

Definition A4.1.3. An **arithmetic sequence** is a sequence where the difference between consecutive terms is a constant.

Definition A4.1.4. A **geometric sequence** is a sequence such that the ratio between consecutive terms is a nonzero constant.

Example 5

Which of the sequences below are arithmetic, which are geometric, which are neither, and what are their common ratios/differences?

- (a) 2, 5, 8, 11, 14, ...
- (b) 3, 6, 12, 24, 48, ...
- (c) 1, 4, 9, 16, 25, ...
- (d) 10, 5, 0, -5, -10, ...
- (e) 2, -2, 2, -2, 2, ...

Example 6

What is the value of $1 + 3 + 5 + \cdots + 99$?

Theorem A4.1.7 (Arithmetic Series)

The sum of the first n terms of an arithmetic sequence, whose first term is a and common difference is d , is

$$a + (a + d) + (a + 2d) + \cdots + (a + (n - 1)d) = \frac{n(2a + (n - 1)d)}{2}$$

Example 8

Compute the sum: $1 + 2 + 4 + \cdots + 32$.

Theorem A4.1.9 (Geometric Series)

The sum of the terms of a geometric sequence, whose first term is a and common ratio is $r \neq 1$, is

$$a + ar + ar^2 + ar^3 + \cdots + ar^n = \frac{a(r^{n+1} - 1)}{r - 1}$$

Infinite Geometric Series

An infinite geometric series only converges when $|r| < 1$.

Example 10

What is the value of $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$?

Theorem A4.1.11 (Infinite Geometric Series)

The sum of the terms of an infinite geometric sequence, whose first term is a and common ratio is r (where $|r| < 1$), is

$$a + ar + ar^2 + ar^3 + \cdots = \frac{a}{1 - r}$$

Recursive Sequences

Definition A4.1.12. A recursive sequence (a_n) is a sequence defined using:

- a **base case**, typically a_0 or a_1 ,
- a **recursive step**, defining a_n in terms of a_{n-1} or other prior terms of the recursive sequence.